

# Source dossier for "Dead Drops on the Blockchain"

The research below consolidates verified, citable sources across all five topic areas requested for the ORF short-form analysis. Every source listed was confirmed through direct web retrieval; where a specific claim could not be substantiated, this is flagged explicitly. The dossier is organised by topic, with full citation details, content summaries, and editorial notes on source quality and gaps.

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## 1. Terror financing via blockchain — South Asian groups and Tron/USDT

This is the empirical backbone of the piece. The sourcing picture is strong for ISIS-affiliated networks using Tron-based USDT in South and Central Asia, but **weaker than expected for JeM and PFI specifically**. The most credible, granular evidence ties ISKP (Islamic State-Khorasan Province) and ISIS modules operating inside India to Tron-based cryptocurrency, not JeM or PFI directly.

### Source 1A — TRM Labs, "Terrorist Financing: Six Crypto-Related Trends to Watch in 2023"

- **Publisher:** TRM Labs Blog
- **Date:** 15 February 2023
- **URL:** <https://www.trmlabs.com/resources/blog/terrorist-financing-six-crypto-related-trends-to-watch-in-2023>
- **Key data:** A **240% increase in USDT use** among tracked terror-financing entities in 2022 versus a 78% rise in BTC. (TRM) Pro-ISIS fundraising in Tajikistan raised ~\$2 million in **Tron-based USDT** for ISKP recruitment. (CoinDesk) A separate Indonesian campaign moved **\$517,000** via USDT on Tron to pro-ISIS channels. (CryptoNews.com) The report also documents ISKP's media arm, al-Azaim Foundation for Media (AFM), soliciting crypto donations from Pakistan-based affiliates. (TRM)
- **Relevance:** The single best source for the Tron/USDT-terror financing nexus in South and Central Asia. Directly supports the argument that stablecoins on high-throughput, low-fee chains are the preferred rail.

### Source 1B — Chainalysis, "Assessing Terrorism Financing On-chain is Crucial and Complex"

- **Publisher:** Chainalysis Blog (part of the 2024 Crypto Crime Report)
- **Date:** 7 May 2024
- **URL:** <https://www.chainalysis.com/blog/assessing-terrorism-financing-on-chain/>
- **Key data:** Documents two primary mechanisms—complex organisational-level financial facilitation (e.g., Hezbollah's networks using MSBs) and small crowdfunding campaigns. Highlights the June 2023 seizure of ~\$1.7 million in cryptocurrency linked to Hezbollah

and Iran's Quds Force by Israel's NBCTF. (chainalysis) Warns that stablecoin use among sanctioned entities and terror financiers is increasing because of challenges accessing USD through traditional channels. (Chainalysis)

- **Relevance:** Provides the analytical framework for how terror organisations interact with blockchain financial infrastructure and the limitations of volume-based assessment.

#### Source 1C — NIA Chargesheet, Mohammad Mohsin Ahmad / ISIS Crypto Financing in India

- **Filed by:** National Investigation Agency (NIA), Government of India
- **Date:** 31 January 2023
- **Coverage:** Business Standard, Republic World, LawBeat, ainvest.com
- **Key data:** NIA charged Ahmad (Patna, Bihar; student at Jamia Millia Islamia) under UAPA Sections 17, 18, 39, and 40 (Devdiscourse) for raising funds for ISIS from Indian and foreign sympathizers, (OneIndia) converting them via a WazirX account, and **sending payments using TRX (Tron) tokens to a Syria-based ISIS-linked online entity called "Ummimmarah."** (The Okhla Times) Ahmad's WazirX account, opened using PAN and Aadhaar, enabled investigators to trace the flow. (Ainvest)
- **Relevance:** A concrete, court-documented Indian case of Tron-based terror financing by an ISIS module operating on Indian soil. Strong evidentiary value.

#### Source 1D — J&K State Investigation Agency, Crypto Terror-Funding Investigation

- **Filed by:** State Investigation Agency (SIA), Jammu & Kashmir Police
- **Date:** 11 July 2025 (raids)
- **Coverage:** The Tribune, Kashmir Observer, DD News (government broadcaster)
- **URLs:** <https://www.tribuneindia.com/news/j-k/3-locations-raided-in-terror-funding-case-involving-cryptocurrency/> ; <https://kashmirobsrver.net/2025/07/11/sia-raids-3-locations-in-jk/>
- **Key data:** SIA Kashmir raided three locations in connection with FIR No. 12/2022, a case involving "a complex conspiracy involving cross-border funding through cryptocurrency, designed to fuel terrorism and destabilise peace in Jammu and Kashmir." (The Tribune +2) The South Asia Terrorism Portal (SATP) reported the investigation focused on **suspicious transactions involving TRX tokens linked to wallets in Pakistan, J&K, and Syria.** (South Asia Terrorism Portal)
- **Relevance:** Directly ties Tron to the Kashmir cross-border terror-financing ecosystem. Does not specifically name JeM, but the geography and modus operandi overlap.

#### Source 1E — T3 Financial Crime Unit (TRON, Tether, TRM Labs)

- **Established:** August 2024

- **Coverage:** Blockworks, Yahoo Finance
- **URL:** <https://blockworks.co/news/tron-tether-and-trm-labs-establish-financial-crime-unit>
- **Key data:** USDT has a \$117+ billion market cap with over 50% of circulating supply on Tron. (Blockworks) Since launch, T3 FCU has frozen over **\$300 million** in criminal assets globally, including assets linked to terrorism financing. (Yahoo Finance) The unit's creation is itself an implicit acknowledgement that USDT on Tron is a preferred rail for illicit actors.
- **Relevance:** Useful for demonstrating the scale of the Tron/USDT problem and industry response.

#### Source 1F — Karnataka High Court, ISIS Member Crypto Bail Denial

- **Court:** Karnataka High Court (Justice HP Sandesh and Justice Venkatesh Naik T)
- **Date:** 2024
- **Coverage:** LiveLaw — <https://www.livelaw.in/high-court/karnataka-high-court/karnataka-high-court-terrorist-activities-crypto-terror-funds-isis-member-520817>
- **Key data:** Court dismissed the bail plea of an accused ISIS member who facilitated transfer of terror funds by sharing cryptocurrency wallet details, received funds from an online handler named "Colonel," and converted crypto to cash.

#### Editorial note on JeM and PFI

**JeM:** The ORF's own publication "From Hawala to Digital Wallets: Jaish-e-Mohammed's Tech-Savvy Terror" (available at orfonline.org) and a VOX-Pol article document JeM's fintech adoption, but the shift is primarily toward **digital wallets** (EasyPaisa, SadaPay, JazzCash), not cryptocurrency. (Wikipedia) JeM's direct crypto use is not well-documented in credible sources. The J&K SIA investigation (Source 1D) is the closest link, but it does not name JeM specifically.

**PFI:** No verified, credible source was found documenting PFI's use of cryptocurrency or USDT for financing. An ANI report from July 2022 about the Phulwari Sharif module actually had Bihar police **refuting** a crypto-funding angle. (ANI News) This is a genuine gap in the public evidentiary record.

**Recommendation for the piece:** Frame the crypto-terror nexus around ISIS-K/ISKP and ISIS modules in India (which are well-documented on Tron/USDT) and the broader Kashmir terror-financing ecosystem, rather than making unsubstantiated claims about JeM or PFI specifically. The ORF's own earlier Expert Speak piece on JeM can be cited for context on JeM's digital evolution.

## 2. On-chain steganography and covert communication via blockchain

This is the strongest section in terms of source availability. A rich, peer-reviewed academic literature exists on blockchain covert channels.

## Source 2A — Partala (2018), "Provably Secure Covert Communication on Blockchain"

- **Author:** Juha Partala
- **Publication:** *Cryptography* (MDPI), Vol. 2, No. 3, Article 18
- **Date:** 20 August 2018
- **DOI:** <https://doi.org/10.3390/cryptography2030018>
- **Summary:** The **foundational paper** in the field. Introduces BLOCCE (Blockchain Covert Communication by Embedding), in which the least significant bits of recipient addresses encode message bits. (Semantic Scholar) Security is proved via "payment indistinguishability" in the random oracle model. (MDPI) Embedding rate is low (~1 bit per block), making it a proof of concept, but it established the theoretical framework for all subsequent work.

## Source 2B — Chen, Zhu, Jiang et al. (2022), "Blockchain Meets Covert Communication: A Survey"

- **Authors:** Zhuo Chen, Liehuang Zhu, Peng Jiang, Can Zhang, Feng Gao, Jialing He, Dawei Xu, Yan Zhang
- **Publication:** *IEEE Communications Surveys & Tutorials*, Vol. 24, No. 4, pp. 2163–2192
- **Date:** 2022
- **DOI:** <https://doi.org/10.1109/COMST.2022.3204281>
- **Summary:** The **most comprehensive survey** in the field (journal impact factor 35.6). Categorises blockchain covert channels into four types: (1) **address channels** — data in LSBs of transaction addresses; (2) **value channels** — encoding in transaction amounts; (3) **DSA/signature channels** — exploiting subliminal channels in ECDSA nonces (kleptographic techniques); (4) **script channels** — OP\_RETURN, smart contract data, input scripts. (ResearchGate) (Wiley Online Library) Also covers transaction filtering, obfuscation, and botnet C2 applications. (Njublockchain)
- **Relevance:** The taxonomic backbone for discussing on-chain steganographic methods in the piece.

## Source 2C — Bartoletti & Pompianu (2017), "An Analysis of Bitcoin OP\_RETURN Metadata"

- **Authors:** Massimo Bartoletti, Livio Pompianu (Università di Cagliari)
- **Publication:** *Financial Cryptography and Data Security* (FC 2017 Workshops), LNCS Vol. 10323, pp. 218–230
- **DOI:** [https://doi.org/10.1007/978-3-319-70278-0\\_14](https://doi.org/10.1007/978-3-319-70278-0_14) ; **arXiv:** <https://arxiv.org/abs/1702.01024>
- **Summary:** The definitive empirical study of OP\_RETURN usage in Bitcoin. OP\_RETURN allows embedding **up to 80 bytes of arbitrary data** per transaction. (Cointelegraph) By February 2017, Bitcoin saw ~25,000 new OP\_RETURN transactions per week (total payload

~323 MB). (Ifca) Critically, ~33% of OP\_RETURN transactions used unknown/unclassified protocols, (Ifca) meaning their purpose was opaque.

#### Source 2D — Tiemann, Berndt, Eisenbarth & Liśkiewicz (2023), "Act natural!": Having a Private Chat on a Public Blockchain"

- **Authors:** Thore Tiemann, Sebastian Berndt, Thomas Eisenbarth, Maciej Liśkiewicz (University of Lübeck)
- **Publication:** 2023 IEEE 8th European Symposium on Security and Privacy (EuroS&P)
- **DOI:** <https://doi.org/10.1109/EuroSP57164.2023.00026> ; **IACR ePrint:** <https://eprint.iacr.org/2021/1073>
- **Summary:** Proposes a public-key subliminal channel embedded in **transaction digital signatures** themselves. Proven undetectable in the random oracle model. Works on any blockchain using ECDSA. (IACR) Any normal-looking transaction can carry covert data — no special fields (like OP\_RETURN) are required. Includes Bitcoin proof-of-concept benchmarks averaged over 1,000,000 runs. (IACR)
- **Relevance:** This is the most operationally alarming method for the piece's argument — it requires no conspicuous fields and is mathematically undetectable.

#### Source 2E — Liu, Fang, Gao et al. (2020), "Whispers on Ethereum"

- **Authors:** Shaoyuan Liu, Zhenshan Fang, Feng Gao, Boris Koussainov, Zijian Zhang, Jianwei Liu, Liehuang Zhu
- **Publication:** Proceedings of the 2nd ACM International Symposium on Blockchain and Secure Critical Infrastructure (BSCI '20), pp. 171–179
- **DOI:** <https://doi.org/10.1145/3384943.3409433>
- **Summary:** The first detailed study targeting Ethereum specifically. Proposes HMAC-based Multiple-Bit Embedding using the VALUE field (transaction amount in Wei) as the carrier. (ACM Digital Library) Ethereum's uint256 value field provides 256-bit embedding capacity per transaction — substantially more than Bitcoin's constrained fields.

#### Source 2F — Alsalami & Zhang (2018/2020), "Uncontrolled Randomness in Blockchains"

- **Authors:** Nasser Alsalami, Bingsheng Zhang (Lancaster University)
- **Publication:** Cryptology ePrint Archive, Report 2018/1184; Lancaster University PhD thesis (2020)
- **ePrint URL:** <https://eprint.iacr.org/2018/1184> ; **Thesis DOI:** <https://doi.org/10.17635/lancaster/thesis/1162>
- **Summary:** Demonstrates that uncontrolled randomness in ECDSA and ring signatures can be exploited for kleptographic and steganographic purposes. (Lancs) Introduces

**Skywhisper**, described as the first blockchain-based broadcast communication tool built on a real cryptocurrency. Also proposes a **Steganography-Resistant Blockchain Framework (SRBF)** as a countermeasure. Key finding: **1.4% of all Bitcoin transactions contain non-financial data**, with over 1,600 files found on-chain. (arXiv) (Lancs)

#### Source 2G — Giron, Martina & Custódio (2021), "Steganographic Analysis of Blockchains"

- **Authors:** Alexandre Augusto Giron, Jean Everson Martina, Ricardo Custódio
- **Publication:** *Sensors* (MDPI), Vol. 21, No. 12, Article 4078
- **DOI:** <https://doi.org/10.3390/s21124078>; **PMCID:** PMC8231769
- **Summary:** Proposes the **first blockchain-specific steganalysis framework** and applies it to real Bitcoin (253 GiB) and Ethereum (107 GiB) data. (PubMed Central) (Academia.edu)  
Crucially, found **no concrete evidence of steganographic use** (ResearchGate +2) — which the authors note means either it is not happening or it is **too difficult to detect**.  
(PubMed Central +2) States that subliminal embeddings in cryptographic primitives are "computationally infeasible for an adversary to detect." (PubMed Central) (MDPI)
- **Relevance:** This is the key source for the "detection gap" argument — even dedicated academic steganalysis cannot find what may be there.

#### Additional reference — MITRE ATT&CK T1102.001

- **Title:** "Web Service: Dead Drop Resolver"
  - **URL:** <https://attack.mitre.org/techniques/T1102/001/>
  - **Summary:** MITRE formally documents the technique of using legitimate web services (including blockchain transactions) as dead drop resolvers for command-and-control communication. (Security Boulevard) Useful as a framing device for the piece's title and concept.
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### 3. Blockchain analytics limitations — the communication-layer blind spot

#### Source 3A — Matzutt et al. (2018), "A Quantitative Analysis of the Impact of Arbitrary Blockchain Content on Bitcoin"

- **Authors:** Roman Matzutt, Jens Hiller, Martin Henze, Jan Henrik Ziegeldorf, Dirk Müllmann, Oliver Hohlfeld, Klaus Wehrle (RWTH Aachen University) (RWTH Publications)
- **Publication:** Proceedings of the 22nd International Conference on Financial Cryptography and Data Security (FC 2018), Springer LNCS
- **URL:** [https://dl.acm.org/doi/10.1007/978-3-662-58387-6\\_23](https://dl.acm.org/doi/10.1007/978-3-662-58387-6_23)

- **Summary:** The seminal empirical study. Found **1.4% of Bitcoin transactions** (3,535,855 transactions, 118.53 MiB of payload) contained non-financial data as of August 2017. (Rwth-aachen) Discovered over **1,600 embedded files**, including objectionable content. (Rwth-aachen) Multiple insertion methods catalogued: OP\_RETURN, coinbase transactions, P2X manipulation, non-standard P2SH scripts. (Roman-matzutt) Standard blockchain analytics tools **do not scan for or flag non-financial data content**.

### Source 3B — Gregoriadis, Muth & Florian (2022), "Analysis of Arbitrary Content on Blockchain-Based Systems using BigQuery"

- **Authors:** Marcel Gregoriadis, Robert Muth, Martin Florian
- **Publication:** *WWW '22 Companion — Proceedings of The Web Conference 2022*, ACM
- **DOI:** <https://doi.org/10.1145/3487553.3524628>
- **Summary:** Extended Matzutt et al.'s work to Bitcoin and Ethereum using Google BigQuery. Found embedded PDF, GIF, SVG, and other file types. Noted significant **false-positive problems**: "we left out file types with very short signatures (e.g., GZIP or DOS executables) because they caused too many false positives." (ACM Digital Library) Detection requires custom tooling — it is not part of any commercial blockchain analytics offering.

### Source 3C — Chen et al. (2024), "Blockchain-Based Covert Communication: A Detection Attack and Efficient Improvement"

- **Authors:** Zhuo Chen, Liehuang Zhu, Peng Jiang, Zijian Zhang, Chengxiang Si (Beijing Institute of Technol...)
- **Publication:** *IEEE Transactions on Information Forensics and Security*, Vol. 19, pp. 9698–9713
- **DOI:** <https://doi.org/10.1109/JIOT.2020.3040389> (Note: verify DOI — may be 10.1109/TIFS.2024.xxxxxxx)
- **URL:** <https://ieeexplore.ieee.org/document/10714428/>
- **Summary:** Demonstrates the detection-evasion **arms race**. Identifies a vulnerability in a state-of-the-art covert channel method (MTMM), then proposes an improved method called ORIM that restores undetectability with **3.7× the bandwidth**. (IEEE Xplore) Key implication: even when detection methods are developed, improved evasion methods quickly emerge. Commercial analytics tools are not engaged in this arms race.

### Source 3D — TU Delft / USENIX Security 2025, Independent Evaluation of Chainalysis

- **Researchers:** TU Delft, Netherlands (associated: Rolf van Wegberg)
- **Presented at:** 34th USENIX Security Symposium, 2025
- **Chainalysis commentary:** <https://www.chainalysis.com/blog/chainalysis-data-independently-proven-accurate-and-reliable/>

- **Summary:** The only controlled independent evaluation of a commercial blockchain analytics vendor. Evaluated Chainalysis against seized server data from three illicit services. Achieved **up to 94.85% true positive rate** with **<0.15% false positive rate** (MEXC) — but exclusively for **financial crime clustering and entity attribution**. Non-financial data, steganographic content, and covert communication were **entirely outside the study's scope** and the tool's design. This is the clearest evidence that even validated analytics tools have zero coverage of the communication layer.

**Source 3E — Zhang, Li, Zhu et al. (2023), "Covert channels in blockchain and blockchain based covert communication"**

- **Authors:** Tao Zhang, Bo Li, Yipeng Zhu, Tong Han, Qianhong Wu
  - **Publication:** *Computer Communications*, Vol. 205, pp. 136-146 (Elsevier)
  - **Date:** 2023
  - **URL:** <https://www.sciencedirect.com/science/article/abs/pii/S0140366423001147>
  - **Summary:** Identifies blockchain as "a natural and ideal channel for helping users establish covert communication channels." States that "there is no systematic analysis on covert channels in blockchain," confirming the gap. Documents covert channels in Bitcoin, Ethereum, Monero, and Zcash (ScienceDirect) at storage and timing layers.
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#### **4. FATF, UN, and government policy documents (2023-2026)**

**Source 4A — FATF, "Targeted Update on Implementation of FATF Standards on Virtual Assets and VASPs" (5th edition)**

- **Organisation:** Financial Action Task Force
- **Date:** 9 July 2024
- **URL:** <https://www.fatf-gafi.org/en/publications/Fatfrecommendations/targeted-update-virtual-assets-vasps-2024.html>
- **Summary:** Finds **75% of 130 assessed jurisdictions** only partially or non-compliant with Recommendation 15. (FATF) Nearly one-third lack Travel Rule legislation. (FATF +2) Notes growing stablecoin use for illicit purposes including terror financing.

**Source 4B — FATF, "Targeted Update on Implementation of FATF Standards on Virtual Assets and VASPs" (6th edition)**

- **Organisation:** Financial Action Task Force
- **Date:** 26 June 2025

- **URL:** <https://www.fatf-gafi.org/en/publications/Fatfrecommendations/targeted-update-virtual-assets-vasps-2025.html>
- **Summary:** Covers 138 jurisdictions. Non-compliance dropped marginally from 25% to 21%. (FATF GAFI) Key finding: **stablecoin use by terror financiers continued to increase**, with most on-chain illicit activity now involving stablecoins. (FATF) Documents DPRK's record \$1.46 billion ByBit theft (FATF) and ~\$51 billion in fraud-related illicit on-chain activity in 2024. (FATF)

#### Source 4C — FATF, "Comprehensive Update on Terrorist Financing Risks"

- **Organisation:** FATF, co-led with UN CTED and France
- **Date:** 8 July 2025
- **URL:** <https://www.fatf-gafi.org/en/publications/Methodsandtrends/comprehensive-update-terrorist-financing-risks-2025.html>
- **Summary:** The most comprehensive FATF report on TF, drawing on 80+ jurisdictions and 840+ submissions. (FATF) Documents "a marked increase in the mixed use of diverse methods of financing and the integration of digital technologies with conventional techniques." (FATF) Finds **69% of assessed jurisdictions** had major or structural deficiencies in effectively investigating/prosecuting TF cases. (FATF) FATF President stated to the UN Security Council (August 2025): "In 2024 ISIL-K has increasingly used virtual assets for organisational transfers and to collect donations internationally." (International Compliance A...)
- **Relevance:** The single most authoritative current source on TF and virtual assets.

#### Source 4D — FATF, "Targeted Report on Stablecoins and Unhosted Wallets — Peer-to-Peer Transactions"

- **Organisation:** Financial Action Task Force
- **Date:** 3 March 2026
- **URL:** <https://www.fatf-gafi.org/en/publications/Virtualassets/targeted-report-stablecoins-unhosted-wallets.html>
- **Summary:** Stablecoin market cap exceeded \$300 billion by mid-2025. Chainalysis data indicates stablecoins accounted for **84% of illicit VA transaction volume in 2025**. (FATF) Highlights P2P transactions via unhosted wallets as a key vulnerability. Discusses misuse by terror financiers, DPRK actors, and drug traffickers. (FATF)

#### Source 4E — UNOCT/UNCCT, Rosfinmonitoring & EAG, "Countering the Misuse of Virtual Assets and VASPs for Terrorism Financing Purposes"

- **Organisation:** UN Office of Counter-Terrorism, UNCCT, Rosfinmonitoring, Eurasian Group
- **Date:** 9 September 2024

- **URL:**  
[https://www.un.org/counterterrorism/sites/www.un.org.counterterrorism/files/uncct\\_cft\\_va\\_report\\_2024\\_en.pdf](https://www.un.org/counterterrorism/sites/www.un.org.counterterrorism/files/uncct_cft_va_report_2024_en.pdf)
- **Summary:** Comprehensive handbook on EAG member states' practices. (United Nations) Covers cryptocurrency risk ratings, mixers/tumblers/anonymisers, and the challenges of VA anonymity. (United Nations) (United Nations) Funded by Russia, Saudi Arabia, and India. (United Nations) (United Nations)

**Source 4F — UNOCT/UNICRI, "Beneath the Surface: Terrorist and Violent Extremist Use of the Dark Web and Cybercrime-as-a-Service for Cyber-Attacks"**

- **Organisation:** UNCCT/UNOCT & UNICRI
- **Date:** June 2024
- **URL:** <https://www.un.org/counterterrorism/en/beneath-surface-terrorist-and-violent-extremist-use-dark-web-and-cybercrime-service-cyber-attacks>
- **Summary:** Examines the convergence of terrorism with cybercrime, including CaaS lowering entry barriers for terrorist groups. (United Nations) (Issuu) Addresses technological infrastructure (including crypto payment rails) enabling this convergence. **The closest UN document to addressing infrastructure/communication-layer exploitation** beyond simple financial transfers.

**Source 4G — UN Security Council CTC, "Algeria Guiding Principles" (S/2025/22)**

- **Organisation:** UN Security Council Counter-Terrorism Committee
- **Date:** 10 January 2025
- **URL:**  
<https://www.un.org/securitycouncil/ctc/sites/www.un.org.securitycouncil.ctc/files/n2500570.pdf>
- **Summary:** Non-binding guiding principles on "preventing, detecting and disrupting the use of new and emerging financial technologies for terrorist purposes." Calls for blockchain analytics in investigations, public-private partnerships, and enhanced VASP oversight. (un)

**Source 4H — CTED, "Insight Briefing: Latest Trends in the Use of Cryptocurrency by Terrorist Groups"**

- **Organisation:** Counter-Terrorism Committee Executive Directorate, UN Security Council
- **Date:** 4 March 2024
- **URL:** <https://www.un.org/securitycouncil/ctc/news/cted-hosts-insight-briefing-latest-trends-use-cryptocurrency-terrorist-groups-and-their>

- **Summary:** Briefing with TRM Labs presenting four key findings: Syria remained a major hub; increasing crypto use by Da'esh affiliates in Asia; stablecoins emerged as TF preference; terrorist groups improving operational security. (United Nations)

#### Source 4I — U.S. Treasury, "2024 National Terrorist Financing Risk Assessment"

- **Organisation:** U.S. Department of the Treasury
- **Date:** February 2024
- **URL:** <https://home.treasury.gov/system/files/136/2024-National-Terrorist-Financing-Risk-Assessment.pdf>
- **Summary:** Documents terrorist groups increasing capability with virtual assets, stablecoin preference, Hamas crypto solicitation, and ISIS converting crypto via hawaladars.

(U.S. Department of the Tre...)

#### Critical gap identified

No FATF, UNODC, or national FIU policy document was found that specifically addresses **communication-layer exploitation of blockchain** — the use of blockchain infrastructure for covert messaging, coordination, or data embedding beyond financial transfers. This is a genuine and significant gap in the policy literature. All existing documents focus overwhelmingly on the financial transfer dimension. This gap itself is a key finding that strengthens the piece's argument: regulators are not even looking at the problem.

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## 5. ORF publication style and format

### Verified editorial guidelines

The ORF's submission guidelines and citation style sheet were directly retrieved.

### Three main formats:

- **Expert Speak:** 800–1,500 words; references via hyperlinks in body text (endnotes in Chicago Style as fallback); reviewed by an expert panel. (Observer Research Founda...)
- **Issue Brief:** 3,000–5,000 words including endnotes; endnotes required (Chicago Manual of Style); double-blind peer review. Tables and graphics encouraged.
- **Occasional Paper:** 5,000–10,000 words including endnotes; endnotes required (CMS); double-blind peer review.

### Citation conventions (from ORF's own style sheet, retrieved at

<https://www.orfonline.org/public/uploads/upload/20240607125834.pdf>):

- Chicago Manual of Style endnotes (not in-text citations) for Issue Briefs and Occasional Papers (orfonline)

- **"Ibid." is NOT used** — shortened note format for repeated citations (orfonline)
- Access dates are NOT included in URLs (orfonline)
- Gender-inclusive and geographically diverse sourcing encouraged (orfonline)
- Authors should cite ORF's own earlier work where relevant (orfonline)
- AI use is strongly discouraged; full disclosure required (orfonline) (Observer Research Founda...)

**Language:** British English throughout — "organisation," "decentralised," "defence," "recognised," "prioritise," "programme."

**Tone:** Analytical, policy-oriented, evidence-based, non-partisan. Descriptive-evaluative rather than prescriptive, with policy recommendations in conclusions.

### Verified example publications directly relevant to the piece's topic

- **Soumya Awasthi, "Exploring the nexus: Cryptocurrency, Zakat, and terror funding,"** ORF Expert Speak, 8 May 2024. (orfonline) URL: <https://www.orfonline.org/expert-speak/exploring-the-nexus-cryptocurrency-zakat-and-terror-funding> — Covers crypto-hawala in India, anonymity challenges, FATF compliance. (ORF Online)
- **Vivek Mishra and Sameer Patil, "Decoding the Biden Administration's Cyber Security Policy,"** ORF Issue Brief No. 686, January 2024. URL: <https://www.orfonline.org/research/decoding-the-biden-administration-s-cyber-security-policy> — Model Issue Brief with 57 endnotes, detailed table, British English throughout.
- **"From Hawala to Digital Wallets: Jaish-e-Mohammed's Tech-Savvy Terror"** — ORF Expert Speak. URL: <https://www.orfonline.org/expert-speak/from-hawala-to-digital-wallets-jaish-e-mohammed-s-tech-savvy-terror> — Directly relevant to JeM's fintech evolution.
- **"PRAHAAR: India's New Counter-Terror Policy in a Hybrid Threat Era"** — ORF Expert Speak. Discusses cryptocurrency tracking, blockchain analysis, terror financing via crypto wallets. (ORF Online) URL: <https://www.orfonline.org/expert-speak/prahaar-india-s-new-counter-terror-policy-in-a-hybrid-threat-era>

**Standard attribution format:** Author(s), "Title," ORF [Publication Type] No. [X], Month Year, Observer Research Foundation.

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### Summary of recommended citation priorities

For a short-form ORF analysis (likely Expert Speak format, 800–1,500 words), the following **minimum citation set** would provide a credible, well-sourced foundation:

| # | Source                                                                                   | What it supports                                            |
|---|------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| 1 | TRM Labs, "Terrorist Financing: Six Crypto-Related Trends" (2023)                        | Tron/USDT as terror financing rail; ISKP data               |
| 2 | FATF, "Comprehensive Update on Terrorist Financing Risks" (2025)                         | Authoritative policy framing; 69% jurisdictional deficiency |
| 3 | Chen et al., "Blockchain Meets Covert Communication: A Survey," <i>IEEE COMST</i> (2022) | Taxonomy of on-chain steganographic methods                 |
| 4 | Tiemann et al., "'Act natural!'," <i>EuroS&amp;P</i> (2023)                              | Undetectable signature-based covert channels                |
| 5 | Matzutt et al., "Arbitrary Blockchain Content," <i>FC 2018</i>                           | 1.4% of Bitcoin transactions contain non-financial data     |
| 6 | FATF, "Stablecoins and Unhosted Wallets" (2026)                                          | 84% of illicit VA volume is stablecoins                     |
| 7 | NIA Chargesheet, Mohsin Ahmad (2023)                                                     | Indian case law: Tron-based ISIS financing                  |

The piece's strongest original contribution will be in identifying the **policy blind spot**: the entire regulatory and analytics ecosystem monitors blockchain's financial layer while the communication layer — where on-chain steganography enables dead-drop-style operational messaging — remains unmonitored, unregulated, and in some cases mathematically undetectable. No existing FATF, UN, or national policy document addresses this gap.